

CMSC 104 Section 2

Spring 2026

Classwork 4

In-class date: March 2, 2026

Due date: before midnight, March 9, 2026

Assignment:

You are going to write a short program that can be used by an adult, 20 years or older, to determine if they are underweight, overweight, or in the “acceptable” zone. Body Mass Index (BMI) is calculated by dividing weight in kilograms by height in meters squared,

Of course, many Americans don’t know off the top of their heads their height in meters, or their weight in kilograms. So you’re going to ask them for their height in feet and inches, and their weight in pounds. And then you’re going to convert to metric units before you calculate BMI.

Take the attached starter code, and complete it, to complete this assignment.

```
#####

# Name:  (insert your name here)

def print_welcome():
    print("Welcome, user. You enter your height in centimeters and your weight
in kilograms")
    print("and we will calculate your Body Mass Index - your BMI")

def get_input(value):

    def convert_units(t, v):
        # write an if/else statement based on the value of t
        # if t == "h" calculate the user's height in meters as v/39.37
        # else calculate the user's weight in kilograms as v/2.2
        # return the calculated value, either height or weight

    if value == "height":
        ft = int(input("Please enter your height in feet and inches. First, how
many feet tall are you?"))
```

```

    inch = ???
    h = 12*ft + inch
    answer = convert_units("h", h)
    print ("You are: ", answer, " meters tall, for future reference")
    return ???
elif value == "weight":
    lbs = int(input("Please enter your weight in pounds. Use whole pounds only;
do not include ounces"))
    answer = convert_units("w", ???)
    print ("You weigh: ", answer, " kilograms, for your future reference")
    return ???

def calc_bmi(height, weight):
    bmi = ???
    return bmi

def print_output(ans):
    print {"Your BMI is: ", ans}
    """
    Now, write an if-elif-else statement that prints the appropriate message:
    - if the user's BMI is under below 18.5, tell the user that they are
underweight
    - else if the user's BMI is between 18.5 and 24.9, inclusive, tell the user
that they are maintaining a healthy weight
    - else if the user's BMI is greater than 24.9 but less than or equal to
29.9, tell the user that they are considered to be overweight
    - else tell the user that they are considered to be obese
    """

if __name__ == "__main__":

    print_welcome()
    meters = get_input("height")
    kg = get_input("weight")
    bmi = calc_bmi (???)
    print_output(???)

```

Reminder

Assignments are your own effort. Do not share your code..

Extra Credit

Use the appropriate code in `print()` to only show two values after the decimal when printing the user's BMI.

Grading Rubric

Script prints:

- Accurate height in meters 25 points
- Accurate weight in kilograms 25 points
- accurate BMI calculation 50 points.
 - Extra Credit: +5 points

What to Submit

Your saved Python script.